Modbus Communication Protocol
EA900 G4 6-20 kVA UPS

Version	Summary of Changes	Date
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1 About the Protocol

1.1 Protocol Description

Modbus protocol is used to make the controller communicate with the monitor via RS485 network.
This communication uses the mode of response. The host initiates requests, and the slave executes requests and responds. The slave is distinguished by the addresses, which ranges from 1 to 247.

1.2 Interface Mode

RS232 interface: asynchronous, half duplex
Baud rate: 9600 bps, settable
Data length: 8 bits, fixed
Parity bit: none, fixed
Stop bit: 1 bit, fixed

1.3 Protocol Format

This protocol supports Modbus RTU mode.

Frame format of RTU mode

When the controller communicates on Modbus in RTU mode, every 8 bits of information contains two 4-bit hexadecimal characters. The format of each byte in RTU mode is as follows:
Coding System: 8-bit binary
Start bit: 1 bit
Data bit: 8-bit data bits, low bit sent first
Parity: 1 bit for odd parity or even parity, 1 stop bit for none parity check
Stop bit: 1 bit
Error checking area: cyclic redundancy check (CRC)

The request frame format of the RTU mode is as follows:

Start	Device address	Function code	Start address of the register	Number of registers	CRC low byte	CRC High byte	End
At least 3.5 characters idle time	8 bit	8 bit	16 bit	16 bit	8 bit	8 bit	At least 3.5 characters idle time

Among them, the character transmission format of RTU mode adopts 10-bit transmission, the data bits are 8 bits. The bit sequence is as follows: (There is no parity bit, so it is 10 bits)

Start bit	1	2	3	4	5	6	7	8	Stop bit (odd/even parity)	Stop bit

The response frame format of the RTU mode is as follows:

Start	Device address	Function code	Data	CRC low byte	CRC High byte	End
At least 3.5 characters idle time	8 bit	8 bit	8n bits	8 bit	8 bit	At least 3.5 characters idle time

A minimum of 3.5 characters' pause interval is required to start the message sending. After the last transmission character, a pause of at least 3.5 characters is required to calibrate the end of the message. A new message can begin after this pause.

The entire message frame must be sent as a continuous stream. If there is more than 1.5 characters idle pause time between the two characters before the frame is completed, frame error is considered, the receiving is stopped and restarted. That is to ensure that the interval between two frames is at least more than 3.5 characters' time. 1.5 characters' time and 3.5 characters' time are related to specific communication baud rate. The calculation method is as follows:
For example, if the communication baud rate is 9600, then
Interval time of 1.5 characters = (1/9600) x 11 x 1.5 x 1000 = 1.72 ms
Interval time of 3.5 characters = (1/9600) x 11 x 3.5 x 1000 = 4.01 ms

【Example】***
Request frame information: Request data from unit No. 1, the position is: register start address being 0002, the number of registers being 1.

	Address	Function code	Start address of the register		Number of registers		CRC check	
Data	0x01	0x03	0x00	0x02	0x00	0x01	0x25	0xCA
Number of bytes	1	1	2		2		2	

Response frame information: Response frame for unit No. 1

	Address	Function code	Number of bytes of data returned	Data content		CRC check	
Data	0x01	0x03	0x02	0x12	0x22	0xE9	0x5C
Number of bytes	1	1	1	2		2	

1.4 Response Information Classification

The host sends a query to the slave and expects a normal response. It may have four events in the host query:

1. The slave receives the query without communication error, and processes information normally, then returns a normal response event.
2. Due to communication error, the slave cannot receive the query data, so it does not return the response. At this time, the host relies on the processor and query timeout are judged.
3. If the slave receives the query, CRC communication error is found and no response is returned, at this time, query timeout is judged by the host processor.
4. The slave receives the query without communication error, when unable to process (such as reading a nonexistent register address or wrong number of registers), it reports the nature of the error to the host.

There are two areas which differ from the normal response when the error response information is reported to the host.
Function code area: In the case of normal response, response function code area for the slave, with the original query function code. MSB for all function codes is 0 (Its value is less than 80H). In the case of abnormal response, MSB of the function code from the slave is set as 1, the value of the function code is made greater than 80H and higher than the value of the normal response. The host application program can identify abnormal response events and can check the data area of abnormal code.
Data area: In a normal response, the data area contains (as required by the query) Data or statistics. In an abnormal response, the data area is an abnormal code that describes the conditions and reasons for the abnormal response from the slave.

The abnormal codes and their meanings are shown in the following table.

Code	Description	Explanation
0x01	Illegal function code	What the slave received is a functional code which cannot be executed. After sending the query command, the code indicates no program function.
0x02	Illegal data address	The data address received is not allowed by the slave, such as register start address error, incorrect number of registers queried.
0x03	Illegal number of registers	Number of registers is out of range.
0x04	Illegal device address	Device address error, no reply required
0x05	Illegal CRC	CRC checking error
0x06	Illegal data length	Data length is out of range.

【Example】***

RTU mode:

Command information: Request data from unit No. 1, the position is: register start address being 0066, the number of registers being 2

	Address	Function code	Start address of the register		Number of registers		CRC check	
Data	0x01	0x03	0x00	0x66	0x00	0x02	0x24	0x14

Response information: Response frame for unit No. 1. The returned message is an illegal data address due to start address of the register error.

	Address	Function code	Data content	CRC check	
Data	0x01	0x83	0x02	0xC0	0xF1

1.5 Function Code

Function code	Description	Explanation
0x02	Read discrete input	Read status information and alarm information
0x03	Read holding registers	Read the current parameter setting information
0x04	Read input registers	Read the current analog amount
0x06	Write a single holding register	Write single setting parameter
0x10	Write multiple holding registers	Write multiple setting parameters

2 Contents of Communication

2.1 Remote Signalling (Function Code 0x02)

1) MCU request command format

	Address	Function code	Start discrete address		Number of discrete		CRC check	
Data	ADDR	02H	High	Low	High	Low	Low	High
Number of bytes	1	1	2		2		2	

2) DSP normal response format

	Address	Function code	Number of bytes of response data	Status of discrete returned	CRC check	
Data	ADDR	02H	DATA_BYTES		Low	High
Number of bytes	1	1	1		2	

3) DSP abnormal response format

	Address	Error code	Exception Code	CRC check	
Data	ADDR	82H	ERR_CODE	Low	High
Number of bytes	1	1	1	2	

Table of remote signalling register

Address	**Description**	**Data type**	**Data length**	**Status meanings**	**Level**
0	BUS high voltage	bit	1	1: occur; 0: disappear	R&D/User
1	BUS low voltage	bit	1	1: occur; 0: disappear	R&D/User
2	BUS unbalanced	bit	1	1: occur; 0: disappear	R&D/User
3	BUS soft start failure	bit	1	1: occur; 0: disappear	R&D/User
4	Inverter soft start failure	bit	1	1: occur; 0: disappear	R&D/User
5	Inverter high voltage	bit	1	1: occur; 0: disappear	R&D/User
6	Inverter low voltage	bit	1	1: occur; 0: disappear	R&D/User
7	Inverter prohibited	bit	1	1: occur; 0: disappear	R&D/User
8	Overtemperature	bit	1	1: occur; 0: disappear	R&D/User
9	Output short circuit	bit	1	1: occur; 0: disappear	R&D/User
10	Overload fault	bit	1	1: occur; 0: disappear	R&D/User
11	Negative power abnormality	bit	1	1: occur; 0: disappear	R&D/User
12	Shutdown fault	bit	1	1: occur; 0: disappear	R&D/User
13	Inconsistent software version for parallel UPSs (Unused)	bit	1	1: occur; 0: disappear	R&D/User
14	Synchronization signal abnormality (Unused)	bit	1	1: occur; 0: disappear	R&D/User
15	Sync pulse abnormality	bit	1	1: occur; 0: disappear	R&D/User
16	Inverter relay adhesion	bit	1	1: occur; 0: disappear	R&D/User
17	BUS short circuit	bit	1	1: occur; 0: disappear	R&D/User
18	CAN communication abnormality	bit	1	1: occur; 0: disappear	R&D/User
19	Physical address conflict	bit	1	1: occur; 0: disappear	R&D/User

20	Parallel UPS model incompatible	bit	1	1: occur; 0: disappear	R&D/User
21	Wrong battery numbers	bit	1	1: occur; 0: disappear	R&D/User
22	Input SCR soft start failure	bit	1	1: occur; 0: disappear	R&D/User
23	Rectifier SCR fault (Unused)	bit	1	1: occur; 0: disappear	R&D/User
24	Inverter abnormality	bit	1	1: occur; 0: disappear	R&D/User
25	Incorrect parallel UPS bypass wiring	bit	1	1: occur; 0: disappear	R&D/User
26	Inverter opencircuit	bit	1	1: occur; 0: disappear	R&D/User
27	PFC abnormality	bit	1	1: occur; 0: disappear	R&D/User
28	Inverter capacitor abnormality	bit	1	1: occur; 0: disappear	R&D/User
29	Fan failed	bit	1	1: occur; 0: disappear	R&D/User
30	EPO	bit	1	1: occur; 0: disappear	R&D/User
31	Auxiliary power lost	bit	1	1: occur; 0: disappear	R&D/User
32	Reserved 1	bit	1	1: occur; 0: disappear	R&D/User
33	Reserved 2	bit	1	1: occur; 0: disappear	R&D/User
34	Reserved 3	bit	1	1: occur; 0: disappear	R&D/User
35	Reserved 4	bit	1	1: occur; 0: disappear	R&D/User
36	Reserved 5	bit	1	1: occur; 0: disappear	R&D/User
37	Reserved 6	bit	1	1: occur; 0: disappear	R&D/User
38	Reserved 7	bit	1	1: occur; 0: disappear	R&D/User
39	Reserved 8	bit	1	1: occur; 0: disappear	R&D/User
40	Reserved 9	bit	1	1: occur; 0: disappear	R&D/User
41	Reserved 10	bit	1	1: occur; 0: disappear	R&D/User
42	Reserved 11	bit	1	1: occur; 0: disappear	R&D/User
43	Reserved 12	bit	1	1: occur; 0: disappear	R&D/User
44	Reserved 13	bit	1	1: occur; 0: disappear	R&D/User
45	Reserved 14	bit	1	1: occur; 0: disappear	R&D/User
46	Reserved 15	bit	1	1: occur; 0: disappear	R&D/User
47	Reserved 16	bit	1	1: occur; 0: disappear	R&D/User
48	Abnormal connection of parallel cables	bit	1	1: occur; 0: disappear	R&D/User
49	Parallel UPS settings error	bit	1	1: occur; 0: disappear	R&D/User
50	Reversed battery connection	bit	1	1: occur; 0: disappear	R&D/User
51	Overload alarm	bit	1	1: occur; 0: disappear	R&D/User
52	Battery disconnected	bit	1	1: occur; 0: disappear	R&D/User
53	Input overcurrent	bit	1	1: occur; 0: disappear	R&D/User
54	High battery voltage	bit	1	1: occur; 0: disappear	R&D/User
55	Startup abnormality	bit	1	1: occur; 0: disappear	R&D/User
56	Charger failure	bit	1	1: occur; 0: disappear	R&D/User
57	EEPROM abnormality	bit	1	1: occur; 0: disappear	R&D/User
58	Input overcurrent overtime	bit	1	1: occur; 0: disappear	R&D/User
59	Low battery voltage	bit	1	1: occur; 0: disappear	R&D/User
60	AD sampling abnormality	bit	1	1: occur; 0: disappear	R&D/User
61	Sync pulse abnormality	bit	1	1: occur; 0: disappear	R&D/User
62	Sync signal abnormality	bit	1	1: occur; 0: disappear	R&D/User
63	CAN communication abnormality	bit	1	1: occur; 0: disappear	R&D/User

64	High bypass voltage	bit	1	1: occur; 0: disappear	R&D/User
65	Bypass fault	bit	1	1: occur; 0: disappear	R&D/User
66	Mains high voltage	bit	1	1: occur; 0: disappear	R&D/User
67	Mains frequency abnormality	bit	1	1: occur; 0: disappear	R&D/User
68	Bypass frequency out of tracking	bit	1	1: occur; 0: disappear	R&D/User
69	Number of switching from bypass to inverter limited	bit	1	1: occur; 0: disappear	R&D/User
70	Battery end-of-discharge	bit	1	1: occur; 0: disappear	R&D/User
71	Battery test success	bit	1	1: occur; 0: disappear	R&D/User
72	Power-on prohibited	bit	1	1: occur; 0: disappear	R&D/User
73	Battery test failed	bit	1	1: occur; 0: disappear	R&D/User
74	Parallel UPS current sharing abnormality	bit	1	1: occur; 0: disappear	R&D/User
75	Inverter phase lock abnormality	bit	1	1: occur; 0: disappear	R&D/User
76	Battery maintenance success	bit	1	1: occur; 0: disappear	R&D/User
77	Battery maintenance abnormality	bit	1	1: occur; 0: disappear	R&D/User
78	Input current unbalanced	bit	1	1: occur; 0: disappear	R&D/User
79	Reserved 16	bit	1	1: occur; 0: disappear	R&D/User
80	PFC abnormality	bit	1	1: occur; 0: disappear	R&D/User
81	Number of switching from Bus low voltage to battery limited	bit	1	1: occur; 0: disappear	R&D/User
82	Mains abnormality	bit	1	1: occur; 0: disappear	R&D/User
83	Bypass abnormality	bit	1	1: occur; 0: disappear	R&D/User
84	Bypass frequency abnormality	bit	1	1: occur; 0: disappear	R&D/User
85	Inconsistent inverter output voltage detection	bit	1	1: occur; 0: disappear	R&D/User
86	Battery voltage abnormality	bit	1	1: occur; 0: disappear	R&D/User
87	Inconsistent bypass output voltage detection	bit	1	1: occur; 0: disappear	R&D/User
88	Instable load (Unused)	bit	1	1: occur; 0: disappear	R&D/User
89	Maintain bypass enabled	bit	1	1: occur; 0: disappear	R&D/User
90	Frequent overtemperature	bit	1	1: occur; 0: disappear	R&D/User
91	Inconsistent number of batteries configured for parallel UPS	bit	1	1: occur; 0: disappear	R&D/User
92	Abnormal battery temperature compensation (Unused)	bit	1	1: occur; 0: disappear	R&D/User
93	Instable bypass voltage (Unused)	bit	1	1: occur; 0: disappear	R&D/User
94	Test mode	bit	1	1: occur; 0: disappear	R&D/User
95	Overload frequently	bit	1	1: occur; 0: disappear	R&D/User

2.2 Telemetering (Function Code 0x04)

1) Request command format of the upper computer

	Address	Function code	Start address of the register		Number of registers		CRC check	
Data	ADDR	04H	High	Low	High	Low	Low	High

Number of bytes	1	1	2	2	2

2) Normal response format

	Address	Function code	Number of bytes of response data	Data returned		CRC check	
Data	ADDR	04H	DATA_BYTES	High	Low	Low	High
Number of bytes	1	1	1	2* number of registers		2	

3) Abnormal response format

	Address	Error code	Exception Code	CRC check	
Data	ADDR	84H	ERR_CODE	Low	High
Number of bytes	1	1	1	2	

Table of telemetering register

Address	**Description**	**Data length**	**Data type**	**Units**	**Factor**	**Level**
0	Bypass phase A voltage	2 bytes	INT16	V	0.1	R&D/User
1	Reserved 2	2 bytes	INT16	/	0.1	R&D/User
2	Reserved 3	2 bytes	INT16	/	0.1	R&D/User
3	Bypass phase A current	2 bytes	INT16	A	0.1	R&D/User
4	Reserved 5	2 bytes	INT16	/	0.1	R&D/User
5	Reserved 6	2 bytes	INT16	/	0.1	R&D/User
6	Bypass frequency	2 bytes	INT16	Hz	0.1	R&D/User
7	Reserved 8	2 bytes	INT16	/	0.1	R&D/User
8	Reserved 9	2 bytes	INT16	/	0.1	R&D/User
9	Bypass PF	2 bytes	INT16	/	0.01	R&D/User
10	Reserved 11	2 bytes	INT16	/	1	R&D/User
11	Reserved 12	2 bytes	INT16	/	1	R&D/User
12	Input phase A voltage	2 bytes	INT16	V	0.1	R&D/User
13	Input phase B voltage	2 bytes	INT16	/	0.1	R&D/User
14	Input phase C voltage	2 bytes	INT16	/	0.1	R&D/User
15	Input phase A current	2 bytes	INT16	A	0.1	R&D/User
16	Input phase B current	2 bytes	INT16	/	0.1	R&D/User
17	Input phase C current	2 bytes	INT16	/	0.1	R&D/User
18	Input frequency	2 bytes	INT16	Hz	0.1	R&D/User
19	Reserved 20	2 bytes	INT16	/	0.1	R&D/User
20	Reserved 21	2 bytes	INT16	/	0.1	R&D/User
21	Input PF	2 bytes	INT16	/	0.01	R&D/User
22	Reserved 23	2 bytes	INT16	/	1	R&D/User
23	Reserved 24	2 bytes	INT16	/	1	R&D/User
24	Output phase A voltage	2 bytes	INT16	V	0.1	R&D/User
25	Reserved 26	2 bytes	INT16	/	0.1	R&D/User
26	Reserved 27	2 bytes	INT16	/	0.1	R&D/User
27	Output phase A current	2 bytes	INT16	A	0.1	R&D/User
28	Reserved 29	2 bytes	INT16	/	0.1	R&D/User
29	Reserved 30	2 bytes	INT16	/	0.1	R&D/User
30	Output frequency	2 bytes	INT16	Hz	0.1	R&D/User

31	Reserved 32	2 bytes	INT16	/	0.1	R&D/User
32	Reserved 33	2 bytes	INT16	/	0.1	R&D/User
33	Output PF	2 bytes	INT16	/	0.01	R&D/User
34	Reserved 35	2 bytes	INT16	/	1	R&D/User
35	Reserved 21	2 bytes	INT16	/	1	R&D/User
36	Output apparent power	2 bytes	INT16	KVA	0.1	R&D/User
37	Reserved 38	2 bytes	INT16	/	0.1	R&D/User
38	Reserved 39	2 bytes	INT16	/	0.1	R&D/User
39	Output active power	2 bytes	INT16	KW	0.1	R&D/User
40	Reserved 41	2 bytes	INT16	/	0.1	R&D/User
41	Reserved 42	2 bytes	INT16	/	0.1	R&D/User
42	Output reactive power	2 bytes	INT16	KVAR	0.1	R&D/User
43	Reserved 44	2 bytes	INT16	/	0.1	R&D/User
44	Reserved 45	2 bytes	INT16	/	0.1	R&D/User
45	Output load percentage	2 bytes	INT16	%	0.1	R&D/User
46	Reserved 47	2 bytes	INT16	/	1	R&D/User
47	Reserved 48	2 bytes	INT16	/	1	R&D/User
48	Reserved 49	2 bytes	INT16	/	1	R&D/User
49	Battery voltage	2 bytes	INT16	V	0.1	R&D/User
50	Reserved 51	2 bytes	INT16	/	1	R&D/User
51	Battery current	2 bytes	INT16	A	0.1	R&D/User
52	Reserved 53	2 bytes	INT16	/	1	R&D/User
53	Battery temperature	2 bytes	INT16	C	0.1	R&D/User
54	Remaining battery discharge time	2 bytes	INT16	Min	1	R&D/User
55	Battery capacity	2 bytes	INT16	%	0.1	R&D/User
56	Inverter current	2 bytes	INT16	A	0.1	R&D/User
57	Rectifier temperature	2 bytes	INT16	C	0.1	R&D/User
58	Inverter temperature	2 bytes	INT16	C	0.1	R&D/User
59	Reserved 60	2 bytes	INT16	/	1	R&D/User
60	Reserved 61	2 bytes	INT16	/	1	R&D/User
61	Reserved 62	2 bytes	INT16	/	1	R&D/User
62	Reserved 63	2 bytes	INT16	/	1	R&D/User
63	Reserved 64	2 bytes	INT16	/	1	R&D/User
64	Reserved 65	2 bytes	INT16	/	1	R&D/User
65	Reserved 66	2 bytes	INT16	/	1	R&D/User
66	Reserved 67	2 bytes	INT16	/	1	R&D/User
67	Software version 1-2	2 bytes	Char	/	1	R&D/User
68	Software version 3-4	2 bytes	Char	/	1	R&D/User
69	Software version 5-6	2 bytes	Char	/	1	R&D/User
70	Status word	2 bytes	Uint16	/	1	R&D/User
71	System operating mode	2 bytes	Uint16	/	1	R&D/User

2.3 Setting Parameters（Function Code 0x06/0x10）

0x06 Command frame format

1) Request command format of the upper computer

	Address	Function code	Register address		Register values		CRC check	
Data	ADDR	06H	High	Low	High	Low	Low	High
Number of bytes	1	1	2		2		2	

2) Normal response format

	Address	Function code	Register address		Register values		CRC check	
Data	ADDR	06H	High	Low	High	Low	Low	High
Number of bytes	1	1	2		2		2	

3) Abnormal response format

	Address	Error code	Exception Code	CRC check	
Data	ADDR	86H	ERR_CODE	Low	High
Number of bytes	1	1	1	2	

Table of command register

Address	**Description**	**Data length**	**Data type**	**Explanation**
0x8000	Clear fault	2 bytes	INT16	1: clear fault; 0: ignore
0x8001	Clear history log	2 bytes	INT16	1: clear history log; 0: ignore
0x8002	Buzzer silence	2 bytes	INT16	1: Buzzer silence; 0: the buzzer works normally
0x8003	Manual switching to bypass	2 bytes	INT16	1: switching to bypass (work in the state of inverter suppling power and normal bypass); 0: ignore
0x8004	Manual switching to inverter	2 bytes	INT16	1: switching to inverter (after manually switching to the bypass power, work when the inverter is normal); 0: ignore
0x8005	Clear battery data	2 bytes	INT16	1: clear battery data; 0: ignore
0x8006	Battery test	2 bytes	INT16	Test type: 1: Battery test (Discharge 20 s) 2: Battery maintenance (discharge until low battery voltage)
0x8007	Stop battery test	2 bytes	INT16	1: Stop battery self-test; 0: ignore

Appendix A Cyclic redundancy check (CRC)

Cyclic redundancy check (CRC) area is 2 bytes, containing a 16-bit binary data. The sending device calculates the CRC value and attaches the value to the message. When the receiving device receives information, it recalculates the CRC value and compares the calculated value with the actual value received in the CRC area, If the two values are different, an error occurs.

CRC set all 16 bits of the register to 1 firstly, then put the adjacent two Data with 8-bit bytes into the current register. Only the 8-bit data of each character is used to generate CRC, the start bit, stop bit and parity bit are not added to CRC.

During generating CRC, every 8 bits of data operates XOR with the values of the register, its result is shifted one bit to the right (in the direction of LSB), and fill in MSB with “0”, detect LSB, if LSB is “1”, XOR with the preset fixed value, if LSB is “0”, not operate XOR.

Repeat the above process until shifting 8 times. After completing the 8th shift, the next 8-bit data XOR with the current value of the register. After all the information has been processed, the final value in the register is the CRC value.